

AVA Business Requirements Document

Business Requirements Document (BRD)

Avakado AI Agent SaaS Platform (AVA_SAAS)

Field	Value
Product	Avakado (AVA) — Multi-tenant AI Agent SaaS
Codebase	ava_saas backend (Node.js / Express / Apollo GraphQL)
Document version	1.0
Generated from	Source code analysis (May 2026)
Primary endpoint	POST /graphql
Public URL	https://www.avakado.ai

1. Executive Summary

Avakado is a **multi-tenant SaaS platform** that enables businesses to build, configure, deploy, and operate **AI-powered conversational agents** across multiple communication channels (web chat, WhatsApp, Telegram, email, SMS, voice, Instagram, Twilio).

The platform combines:

- **Knowledge management** (RAG) via collections sourced from websites, files, YouTube, and plain text
- **Custom agent configuration** (prompts, models, appearance, voice, actions, workflows)
- **Omnichannel deployment** with webhook-driven inbound messaging
- **Outbound campaigns** and scheduled jobs for telephony (Twilio, Exotel, Tata Tele)
- **CRM-style lead management** with templates and bulk import
- **Subscription billing** via Razorpay with credit-based usage (LLM, knowledge, miscellaneous)
- **Third-party integrations** (Zoho CRM, Twilio Connect, OAuth providers)
- **Support ticketing** and in-app notifications

This BRD is derived from the implemented backend (`src/`) and reflects **what the system does today**, not a future roadmap.

2. Business Objectives

ID	Objective	Success Criteria
BO-01	Enable businesses to deploy branded AI agents without building infrastructure	Tenant can register, create agent, attach knowledge, and go live on ≥ 1 channel
BO-02	Centralize knowledge ingestion and retrieval for accurate agent responses	Collections process website/file/YouTube sources into embedded chunks
BO-03	Support omnichannel customer engagement	Agents operate on web, messaging, email, SMS, and voice channels
BO-04	Monetize via subscription plans and usage credits	Plans, subscriptions, and per-business credit balances enforced
BO-05	Provide outbound sales/support automation	Campaigns schedule outbound calls with agent context
BO-06	Integrate with external CRM and telephony systems	Zoho, Twilio, Exotel, Tata Tele connectors available
BO-07	Maintain tenant isolation and role-based access	All tenant data scoped to <code>business</code> ; scopes gate GraphQL operations

3. Stakeholders & User Roles

3.1 Stakeholders

Stakeholder	Interest
Business tenant (customer)	Configure agents, knowledge, channels, campaigns; view analytics
Platform operator (Avakado)	Manage plans, super-admin operations, marketplace templates
End user (consumer)	Interacts with deployed agents via channels (no direct platform login)
QA / Engineering	API contracts, test fixtures, integration verification

3.2 Platform User Roles

Role	Description	Default Capabilities
superAdmin	Platform-level administrator	<code>super:all</code> and full admin scopes
admin	Business owner / primary administrator	Full CRUD on agents, collections, channels, integrations, workflows, billing upgrade
manager	Limited operational user	Read-heavy: conversations, analytics, tickets; limited create on tickets

Roles are assigned at registration (`admin` for business creator) or via `createUser` mutation.

4. Functional Requirements

4.1 Authentication & Onboarding

ID	Requirement	Implementation
FR-AUTH-01	User registration creates User + Business atomically	<code>register</code> mutation; MongoDB transaction
FR-AUTH-02	Email verification required before login	<code>isVerified</code> flag; verification email via <code>WEBHOOKS_URL</code>
FR-AUTH-03	Login returns JWT access token, role, scopes, user profile	<code>login</code> mutation; cookies supported via middleware
FR-AUTH-04	Scope-based authorization on GraphQL fields	Directives: <code>@requireScope</code> , <code>@requireBusinessAccess</code> , <code>@requireResourceOwnership</code>
FR-AUTH-05	Public operations without auth	<code>login</code> , <code>register</code> , <code>ephemeralToken</code> , <code>fetchPublicPlans</code> , introspection

Business rules:

- Duplicate business name or email rejected at registration
- Password hashed with bcrypt (12 rounds)
- Access/refresh tokens: JWT, default 30-day expiry

4.2 Business (Tenant) Management

ID	Requirement	Details
FR-BIZ-01	Each user belongs to one business	User.business → Business
FR-BIZ-02	Business profile includes branding and contact	name, logoURL, sector, tagline, facts, quickQuestions, address, description, contact
FR-BIZ-03	Credit wallet per business	llmCredits , knowledgeCredits , miscellaneousCredits , balance , active subscription plan
FR-BIZ-04	Usage analytics tracked	engagement overview, conversation analytics, credit usage breakdown (chat, knowledge, analysis)
FR-BIZ-05	Conversation date retention	MAX_DAYS default 45; pruning of old daily counts

4.3 AI Agent Management

ID	Requirement	Details
FR-AGENT-01	Create/update/delete agents per business	CRUD mutations with agent:* scopes
FR-AGENT-02	Agent configuration	appearance (color boxes), personalInfo (name, systemPrompt, quickQuestions, welcomeMessage, model, temperature, VoiceAgentSessionConfig)
FR-AGENT-03	Agent associations	collections[], workflow, channels[], actions[]
FR-AGENT-04	Public/featured agents	isPublic , isFeatured flags for marketplace
FR-AGENT-05	Voice realtime sessions	ephemeralToken (OpenAI or Gemini provider) — public operation
FR-AGENT-06	Prompt generation helper	generatePrompt mutation for AI-assisted prompt building
FR-AGENT-07	Analysis metrics	analysisMetrics JSON on agent

Supported voice models (from schema): OpenAI realtime models, voices (alloy, ash, coral, etc.), turn detection (server_vad, semantic_vad).

4.4 Knowledge Collections (RAG)

ID	Requirement	Details
FR-COLL-01	Collection types by source	<code>website</code> , <code>youtube</code> , <code>file</code> , <code>text</code>
FR-COLL-02	Processing status lifecycle	<code>loading</code> → <code>active</code> or <code>failed</code>
FR-COLL-03	Website ingestion	Firecrawl batch scrape; webhook callback on completion
FR-COLL-04	File ingestion	Presigned upload to Cloudflare R2 (<code>ava-client-documents</code>); LlamaParse parsing via <code>source_url</code>
FR-COLL-05	Chunking strategies	<code>recursiveStructural</code> , <code>recursiveSemantic</code> with tunable JSON
FR-COLL-06	Parser tiers	<code>fast</code> , <code>cost_effective</code> , <code>agentic</code> , <code>agentic_plus</code>
FR-COLL-07	Parser output expansion	<code>text</code> , <code>items</code> , <code>markdown</code> , <code>metadata</code> , <code>images_content_metadata</code> , <code>xlsx_content_metadata</code> , <code>output_pdf_content_metadata</code>
FR-COLL-08	File storage operations	<code>getUploadUrl</code> , <code>getDownloadUrl</code> , <code>deleteUploadedFileFromStorage</code> , <code>getListOfUploadedFiles</code>
FR-COLL-09	Direct upload (Multer)	PDF and TXT up to 50MB (middleware exists; route wiring optional)

Business rule: Knowledge processing consumes `knowledgeCredits`; embeddings and summarization costs tracked in business analytics.

4.5 Custom Actions (Agent Tools)

ID	Requirement	Details
FR-ACT-01	Define executable JavaScript functions	<code>functionString</code> , <code>errorFunction</code>

ID	Requirement	Details
FR-ACT-02	Parameter schema	parameters JSON; config JSON
FR-ACT-03	Execution controls	async , needsApproval flags
FR-ACT-04	UI rendering config	UI JSON for action presentation
FR-ACT-05	Public actions	isPublic for marketplace sharing

4.6 Communication Channels

ID	Requirement	Channel Types
FR-CHAN-01	Create channel per type	email, whatsapp, telegram, web, phone, sms, instagram, twilio
FR-CHAN-02	Channel configuration	config JSON, systemPrompt , UIElements , optional workflow link
FR-CHAN-03	Webhook URL per channel	Auto-generated for inbound message routing
FR-CHAN-04	OAuth integrations	Google, Microsoft, Instagram, WhatsApp, Twilio Connect under src/services/oauth/

4.7 Conversations & Messages

ID	Requirement	Details
FR-CONV-01	Store conversation sessions	Linked to agent, channel, optional campaign, workflow state
FR-CONV-02	Conversation statuses	initiated, active, interrupted, inactive, disconnected, noAnswer, busy, failed, completed, inProgress
FR-CONV-03	Metadata tracking	totalMessages, reactions (like/dislike/neutral), userLocation, socket info, browserUrl

ID	Requirement	Details
FR-CONV-04	Transcripts & extracted data	<code>transcripts</code> , <code>extractedData</code> JSON fields
FR-CONV-05	Message history	<code>fetchMessages</code> by <code>conversationId</code> with pagination
FR-CONV-06	Message reactions (REST)	<code>PUT /reaction</code> — neutral, like, dislike

Note: Conversations and messages have **queries only** (no GraphQL mutations); updates occur via webhooks and internal services.

4.8 Workflows

ID	Requirement	Details
FR-WF-01	Visual workflow graphs	<code>nodes</code> and <code>connections</code> JSON
FR-WF-02	Inbuilt node library	Reusable node templates (ports, core, meta)
FR-WF-03	Test individual nodes	<code>testWorkflowNode</code> mutation
FR-WF-04	Channel/workflow binding	Channels and agents can reference workflows

4.9 Campaigns & Scheduled Jobs

ID	Requirement	Details
FR-CAMP-01	Campaign entity	name, agent, <code>communicationChannels[]</code> , <code>leads[]</code> , <code>schedule</code> , <code>cps</code> , <code>instructions</code> , <code>nodes/edges</code>
FR-CAMP-02	Job scheduling	External Bull/Agenda service via <code>BULL_URL</code>
FR-CAMP-03	Job types	<code>outboundCall</code> with payload (channel, agent, to, <code>PreContext</code> , <code>retries</code> , <code>CPS</code>)
FR-CAMP-04	Job statuses	<code>scheduled</code> , <code>active</code> , <code>completed</code> , <code>failed</code> , <code>canceled</code> , <code>delayed</code> , <code>waiting</code>
FR-CAMP-05	Immediate outbound call	<code>makeAnOutboundCall</code> mutation

ID	Requirement	Details
FR-CAMP-06	Exotel campaign setup	<code>exotelCampaignSetup</code> with contacts and schedule
FR-CAMP-07	Tata Tele testing	<code>testTataTele</code> mutation

4.10 Lead Management (CRM)

ID	Requirement	Details
FR-LEAD-01	Lead templates with dynamic fields	<code>fields</code> JSON schema per template
FR-LEAD-02	Lead records	<code>data</code> JSON conforming to template; notes, tags, status, origin
FR-LEAD-03	Bulk import	<code>bulkCreateLeads</code> mutation
FR-LEAD-04	Faceted filtering	<code>fetchLeadFacets</code> for filter UI

4.11 Integrations

ID	Type	Purpose
FR-INT-01	zoho	CRM OAuth, WorkDrive file upload
FR-INT-02	twilio	SMS, voice, phone numbers, Connect app
FR-INT-03	exotel	India telephony campaigns
FR-INT-04	tataTele	Tata Telephony testing
FR-INT-05	whatsapp	WhatsApp Business API

Operations: `createIntegration` , `fetchIntegration` , `deAuthorizeIntegration` .

4.12 Twilio Operations (Extended)

Category	Capabilities
Account	Account details, balance
Phone numbers	List available, list owned, buy, update, release
Messaging	Send SMS/MMS, list messages, SMS status

Category	Capabilities
Voice	Outbound test call, AI outbound call with agent context
Recordings	List call recordings, transcriptions
Usage	Usage records (calls, SMS), timely usage, pricing by country

4.13 Service Provider Marketplace

ID	Requirement	Details
FR-SP-01	External API provider registry	Providers with basicScopes, icon, color
FR-SP-02	API definitions	schemas (input/config/output/error), requestTemplate, auth types
FR-SP-03	Authenticators	OAuth2, API key, bearer, JWT, HMAC, mTLS, etc.
FR-SP-04	Auth strategies	<code>createAuthStrategy</code> , <code>createApiAuthenticator</code>

4.14 Billing & Subscriptions

ID	Requirement	Details
FR-PAY-01	Plan types	FREE, BASE, TOPUP
FR-PAY-02	Plan credits	llm, knowledge, miscellaneous per plan
FR-PAY-03	Public plan listing	<code>fetchPublicPlans</code> (no auth)
FR-PAY-04	Subscription lifecycle	<code>startPayment</code> (Razorpay), <code>cancelSubscription</code>
FR-PAY-05	Super-admin plan management	<code>createAVAPlan</code> , <code>updateAVAPlan</code> , <code>deleteAVAPlan</code>

4.15 Support Tickets

ID	Requirement	Details
FR-TKT-01	Ticket creation (REST)	<code>POST /raise-ticket</code> — business, issueSummary, channel, priority, contactDetails

ID	Requirement	Details
FR-TKT-02	Ticket management (GraphQL)	fetch, update with email response, delete
FR-TKT-03	Priorities	low, medium, high
FR-TKT-04	Statuses	pending, responded, resolved

4.16 Notifications

ID	Requirement	Details
FR-NOT-01	In-app notifications	head, body, type, attachments JSON
FR-NOT-02	Seen/unseen status	update and delete operations

4.17 REST Utility Endpoints

Endpoint	Method	Purpose
/	GET	Health check
/reaction	PUT	Message reaction update
/send-invite	POST	iCal meeting invites (AVA or custom SMTP)
/send-mail	POST	Outbound email
/contact-us	POST	Contact form to internal team
/raise-ticket	POST	Create support ticket
/webhook/firecrawl/*	*	Crawl completion callbacks
/webhook/telegram/*	*	Telegram bot events
/webhook/whatsapp/*	*	WhatsApp inbound
/webhook/instagram/*	*	Instagram inbound

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	Helmet headers, mongo-sanitize on inputs, CORS whitelist, JWT auth

ID	Category	Requirement
NFR-02	Multi-tenancy	@requireBusinessAccess enforces business-scoped data
NFR-03	API format	GraphQL responses wrapped: { success, message, data }
NFR-04	Payload size	JSON body limit 50MB
NFR-05	Observability	Morgan HTTP logging; GraphQL error logging
NFR-06	Real-time	Socket.io client for status updates to frontend rooms
NFR-07	Async processing	External job queue (BULL_URL) for URL processing and job sync
NFR-08	Scalability	Stateless API; MongoDB persistence; R2 object storage

6. Data Model Summary

Entity	Key Relationships
Business	owns agents, collections, channels, credits, analytics
User	belongs to business; has role and scopes
Agent	→ collections, workflow, channels, actions, business
Collection	→ business; content sources with processing status
Channel	→ business, workflow; type-specific config
Conversation	→ agent, channel, campaign, workflow
Message	→ conversation, business
Action	→ business; JS function tools
Workflow	→ business; nodes/connections graph
Campaign	→ agent, channels, leads
Job	→ campaign; scheduled outbound calls
Lead / LeadTemplate	→ business
Integration	→ business; OAuth tokens in config
Plan / Subscription	→ business credits
Ticket	support requests

Entity	Key Relationships
Notification	user notifications

7. External Dependencies

Service	Usage
MongoDB	Primary database
OpenAI	Chat, embeddings, summarization, realtime voice
Google Gemini	Alternative voice provider
LlamaCloud / LlamaParse	Document parsing
Firecrawl	Website scraping
Cloudflare R2	Document storage (<code>ava-client-documents</code>)
Twilio	SMS/voice telephony
Exotel / Tata Tele	India telephony
Razorpay	Payment gateway
Zoho CRM	CRM integration
WhatsApp / Telegram / Instagram APIs	Channel webhooks
LocationIQ	Reverse geocoding
Nodemailer	Email delivery
WEBHOOKS_URL service	Email verification, auxiliary triggers
BULL_URL	Job queue worker

8. Authorization Model

8.1 Directives

- `@requireScope(scope)` — user must have scope or `super:all`
- `@requireAnyScope(scopes)` — any one scope
- `@requireAllScopes(scopes)` — all scopes
- `@requireBusinessAccess` — operation limited to user's business tenant
- `@requireResourceOwnership` — creator or business ownership check

8.2 Scope Categories (abbreviated)

Authentication, business, agent, collection, channel, action, conversation, ticket, analytics, file, template, admin, super, webhook, subscription, notification, integration, workflow.

Full scope list and role mappings are defined in `src/models/User.js` (`ScopesEnum` , `RoleScopes`).

9. User Journeys

9.1 Business Onboarding

```
flowchart LR
  A[Register] --> B[Verify Email]
  B --> C[Login]
  C --> D[Create Collection]
  D --> E[Create Agent]
  E --> F[Create Channel]
  F --> G[Deploy / Go Live]
```

9.2 Inbound Conversation

```
flowchart LR
  W[Webhook Event] --> C[Create/Update Conversation]
  C --> M[Store Messages]
  M --> A[Agent LLM + RAG]
  A --> R[Response to Channel]
```

9.3 Outbound Campaign

```
flowchart LR
  L[Import Leads] --> CP[Create Campaign]
  CP --> J[Schedule Jobs]
  J --> T[Telephony Integration]
  T --> CV[Conversation + Transcript]
```

10. Constraints & Assumptions

Item	Detail
CORS	Whitelisted origins only (avakado.ai, localhost, Apollo Studio)
Zoho GraphQL module	Present in codebase but not mounted in active schema

Item	Detail
File upload route	Multer middleware exists; not wired to Express routes in current <code>server.js</code>
Manager role	Read-heavy; cannot create agents or collections
Credits	Plan activation and credit deduction logic tied to business model methods

11. Out of Scope (Current Implementation)

- Native mobile SDK
- Multi-business membership per user
- GraphQL mutations for conversations/messages (webhook-driven only)
- Password reset flow (`forgotPassword` listed public but resolver commented)
- Zoho-specific GraphQL module (disabled)

12. Acceptance Criteria (Platform-Level)

#	Criterion
1	New business can register and receive verification email
2	Verified admin can login and receive scoped JWT
3	Admin can create collection from website URL and file upload path
4	Admin can create agent bound to collection and channel
5	Inbound webhook creates conversation and stores messages
6	Admin can view conversations and messages with pagination
7	Admin can configure Twilio integration and place test outbound call
8	Admin can create campaign with leads and schedule outbound job
9	Public plans visible without authentication
10	Authenticated admin can start Razorpay subscription
11	Support ticket can be raised via REST and managed via GraphQL
12	All tenant queries return only business-scoped data

13. Glossary

Term	Definition
Agent	Configured AI assistant with prompt, model, knowledge, and channel bindings
Collection	Knowledge base container with one or more content sources
Channel	Communication endpoint (web widget, WhatsApp number, etc.)
Action	Custom tool/function the agent can invoke
Workflow	Visual automation graph attached to agents or channels
Campaign	Outbound outreach configuration with leads and schedule
Job	Scheduled task (e.g., outbound call) processed by external queue
Scope	Fine-grained permission string gating GraphQL operations
Credits	Usage units for LLM, knowledge processing, and misc operations

This document was auto-generated from the Ava_SAAS codebase. For API operation details, see `docs/graphql/AVA_GraphQL_API_Reference.md`. For test artifacts, see `test-fixtures/`.